The Even Icosahedral Case of Artin's Conjecture: Certificates, Certified Pole Bounds, and the Missing Maass Form

A computational survey note

Abstract

We report a computational study of the even icosahedral case of Artin's holomorphy conjecture — totally real quintic fields with Galois group A_5, for which not a single instance is known.

Algebraic. A linear program over induced linear characters gives the exact *monomial defect* of every irreducible character of the relevant groups. For the three-dimensional character of A_5 the defect is $1/4$, with an explicit optimal certificate showing that a pole of order m of $L(s, \rho_3)$ forces a zero of order $\geq 4m$ of a degree-60 Dedekind zeta function. The value $1/2$ occurs only in Schur double covers.

Arithmetic. The two A_5-classes of 5-cycles are provably invisible to every subfield factorisation. We implement the Vandermonde-sign resolution and record a methodological warning: a Chebotarev census *cannot* validate it. We derive a local-sign rule for ramified Euler factors from the geometry of $A_5 \subset SO(3)$, validated three for three, and a method for recovering Euler factors at index primes.

Analytic. We give a *residue-independent* pole test based on the Weil explicit formula, in which a pole contributes a full unit regardless of its residue. We then give a fully certified bound $|D(1.10)| \leq 4.5 \times 10^{-12}$ for the minimal even icosahedral case (Doud–Moore, conductor 1951), computed in ball arithmetic with *every* error term proven.

Geometric. We construct an explicit $\lambda = 1/4$ Maass form of level 229 from root counts of a cubic and verify $\Gamma_0(229)$-invariance to 2×10^{-13}, then show that the obstruction to doing the same in the icosahedral case is the absence of an index-2 subgroup of A_5 — the same one-line fact as the monomiality obstruction, in geometric clothing.

We also record what failed, including two survey fields whose discrepancy survives every internal check, and seven claims made and retracted during the work (§14).

Contents

1 Introduction

Artin's conjecture [1] asserts that $L(s,\rho)$ is entire for every nontrivial irreducible complex representation ρ of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Brauer's induction theorem [2] gives meromorphic continuation and the functional equation unconditionally; holomorphy is what remains.

For two-dimensional representations the state of the art is sharply asymmetric.

- **Odd** ($\det\rho(c) = -1$): a theorem, via Khare–Wintenberger [4] with Kisin's modularity lifting, building on [3].
- **Even** ($\rho(c) = \pm I$): open. Langlands and Tunnell cover the tetrahedral and octahedral cases by solvable base change; A_5 is simple, base change fails, and even representations correspond to Maass forms of eigenvalue $\lambda = 1/4$, which carry no geometric Galois

representation for deformation-theoretic machinery to act on. As Calegari observes [5], not a single even icosahedral case is known.

The three-dimensional irreducible ρ_3 of A_5 furnishes a parallel family. For a *non*-totally-real A_5 quintic, holomorphy of $L(s, \rho_3)$ follows: the associated two-dimensional ρ is odd, hence modular, and $\mathrm{Sym}^2 \rho$ is automorphic by Gelbart–Jacquet. For a *totally real* A_5 quintic no such route exists.

Here ρ_3 has a decisive computational advantage: since the symmetric square annihilates the centre of $\mathrm{SL}(2,5)$, $\rho_3 \cong \mathrm{Sym}^2 \rho \otimes (\det \rho)^{-1}$ is insensitive to the choice of lift. The embedding problem $2.A_5 \to A_5$, which obstructs direct computation with ρ (cf. Buhler [6]), disappears entirely.

We note a point often stated incorrectly: the smallest non-monomial group is $\mathrm{SL}(2,3)$, of order **24**, not A_5. Monomiality and solvability are independent — $\mathrm{SL}(2,3)$ is solvable and non-monomial, while the 5-dimensional character of the simple group A_5 *is* monomial (induced from a cubic character of A_4).

2 The monomial defect

2.1 Definition and computation

For a finite group G and $\chi \in \mathrm{Irr}(G)$, let $\mathcal{M}$ be the set of characters $\mathrm{Ind}_H^G \lambda$ with $H \le G$ and λ linear. Define

$$\mathrm{def}(\chi) = \min \Big\{ \textstyle\sum_j b_j \;:\; \chi = \sum_i a_i \sigma_i - \sum_j b_j \tau_j,\; \sigma_i, \tau_j \in \mathcal{M},\; a_i, b_j \in \mathbb{Q}_{\ge 0} \Big\}.$$

This is a rational linear program over finitely many generators, hence exactly computable. Since $L(s,\sigma)$ is an abelian Hecke L-function for every $\sigma \in \mathcal{M}$, the defect bounds how badly poles can occur.

G	defects by degree	max
$S_3,\ A_4,\ S_4$, all $\mathbb{Z}/q \rtimes \mathbb{Z}/(q-1)$	all 0	0
A_5	$1{:}0,\ 3{:}\frac{1}{4},\ 3'{:}\frac{1}{4},\ 4{:}\frac{2}{5},\ 5{:}0$	2/5
A_6	$1{:}0,\ 5{:}\frac{1}{19},\ 5'{:}\frac{1}{19},\ 8{:}\frac{1}{3},\ 8'{:}\frac{1}{3},\ 9{:}\frac{3}{7},\ 10{:}0$	3/7
$\mathrm{PSL}(2,7)$	$3{:}\frac{1}{4},\ 3'{:}\frac{1}{4},\ 6{:}\frac{1}{3},\ 7{:}0,\ 8{:}0$	1/3
$\mathrm{SL}(2,3) = 2T$	$2{:}\frac{1}{2}$ $(\times 3)$, $3{:}0$	1/2
$2O$ (binary octahedral)	faithful $2{:}\frac{1}{2}$ $(\times 2)$, all others 0	1/2
$\mathrm{GL}(2,3) = 2.S_4^-$	$2{:}\frac{1}{2}$ $(\times 2)$	1/2
$\mathrm{SL}(2,5) = 2I$	$2{:}\frac{1}{2},\ 2'{:}\frac{1}{2},\ 3{:}\frac{1}{4},\ 4{:}\frac{1}{2},\ 4'{:}\frac{2}{5},\ 5{:}0,\ 6{:}0$	1/2
$\mathrm{SL}(2,7)$	$3{:}\frac{1}{4},\ 4{:}\frac{1}{6},\ 6{:}\frac{1}{2},\ 7, 8{:}0$	1/2

No defect exceeds 1/2 in any group computed, and 1/2 occurs **only** in Schur double covers: the simple groups stay strictly below ($A_5 \to 2/5$, $A_6 \to 3/7$, $\mathrm{PSL}(2,7) \to 1/3$), while the covers attain it, and within each cover only on the *faithful* (spin) characters.

We stress what the defect does *not* measure. $\mathrm{SL}(2,3)$ (Langlands) and $2O$ (Tunnell) are **proved** cases carrying the same defect 1/2 as the open icosahedral one. The defect quantifies failure of the positive-induction argument, not difficulty of the problem.

2.2 The certificate for χ_3

The LP optimum for χ_3 is attained at

$$4\chi_3 = 4\, \mathrm{Ind}_{C_5}^{A_5} \lambda_0 + \mathrm{Ind}_{C_5}^{A_5} \mathbf{1} + 2\, \mathrm{Ind}_{H_{10}}^{A_5} \mu - \mathrm{reg}_{A_5},$$

with λ_0 a nontrivial character of a cyclic C_5 and μ a nontrivial linear character of a subgroup of order 10. Degrees: $4(12)+12+2(6)-60=12=4\cdot 3$. Translating to L-functions,

$$L(s,\rho_3)^4\,\zeta_L(s)=L(s,\lambda_0,K_{12})^4\,\zeta_{K_{12}}(s)\,L(s,\mu,K_{10})^2.$$

Proposition 2.1. *If $L(s,\rho_3)$ has a pole of order m at $s_0\neq 1$, then* $\mathrm{ord}_{s_0}\zeta_L\geq 4m$.

This is a sharp instance of the Foote–Murty inequality [7]; the phenomenon is the subject of Foote–Wales [8]. What is new is only that the LP returns the *optimal* constant.

2.3 A correction: $\mathrm{GL}(2,3)$ is not the binary octahedral group

Every finite subgroup of the unit quaternions has exactly one involution (the central -1).

group	order	involutions	binary polyhedral?
$\mathrm{SL}(2,3)$	24	**1**	yes — $2T$
$\mathrm{GL}(2,3)$	48	**13**	**no**
$\mathrm{SL}(2,5)$	120	**1**	yes — $2I$

$\mathrm{GL}(2,3)$ is $2.S_4^-$, the *other* double cover of S_4. The genuine $2O$ has order 48, one involution, and the same character degrees $[1,1,2,2,2,3,3,4]$ — which is how the confusion arises. We had mislabelled it through an earlier draft.

With the correct group, a $(2,3,n)$ pattern holds across the **complete** binary polyhedral family: an optimal defect-1/2 certificate has positive part induced from C_{2n} and negative part from C_6:

$$\begin{aligned}
n=3:&\quad \mathrm{Ind}\,\lambda_0+\mathrm{Ind}\,\lambda_1-\mathrm{Ind}\,\lambda_4=2\chi\quad \text{(three characters of one } C_6)\\
n=4:&\quad 2\,\mathrm{Ind}_{C_8}\lambda-\mathrm{Ind}_{C_6}\mu=2\chi,\qquad 2(6)-8=4\\
n=5:&\quad 2\,\mathrm{Ind}_{C_{10}}\lambda-\mathrm{Ind}_{C_6}\mu=2\chi
\end{aligned}$$

These exhaust the non-cyclic, non-dihedral finite quaternion subgroups. A previously recorded "refutation" via $\mathrm{SL}(2,7)$ was misapplied: $\mathrm{SL}(2,7)$ is not binary polyhedral.

2.4 Where non-monomiality begins

A computation over all subgroups of $\mathrm{GL}(2,3)$ locates the first failure: every group of order 2^a3^b below 24 has defect 0, as does S_4 (order 24), and the first nonzero defect appears at $\mathrm{SL}(2,3)$, of order $2^3\cdot 3$.

Burnside's p^aq^b theorem forbids a non-abelian simple group of order 2^a3^b. Hence:

> Inside the 6-smooth world, **solvability is guaranteed but monomiality is not.** Non-solvability requires a third prime, and the smallest available is 5.

This aligns with the Platonic boundary: $|2T|=2^3\cdot 3$ and $|2O|=2^4\cdot 3$ are 6-smooth; $|2I|=2^3\cdot 3\cdot \mathbf{5}$ is not. The character fields line up too — the faithful 2-dimensional character takes the value $2\cos(\pi/n)$ on elements of order $2n$, so its real character field is $\mathbb{Q}(\zeta_{2n})^+$, giving $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$ for $n=3,4,5$. The full character field of $2T$ is $\mathbb{Q}(\zeta_3)$, whose ring of integers is the Eisenstein integers.

Remark. A mechanism we tested and *rejected*: "quaternionic type plus rational values forces non-monomiality" is false — the 6-dimensional of $\mathrm{SL}(2,5)$ is quaternionic, rational-valued, and *is* a single induced character. Monomiality gives Schur index 1 over $\mathbb{Q}(\lambda)$, not over $\mathbb{Q}(\chi)$. What survives is the index-2 argument: $\mathrm{SL}(2,3)$ has $G^{\mathrm{ab}}=C_3$ and $\mathrm{SL}(2,5)$ is perfect, so neither has a subgroup of index 2, hence neither has a degree-2 monomial character. This one line covers both, and reappears geometrically in §10.4.

3 Separating the two classes of 5-cycles

Lemma 3.1. *No subfield of L has a factorisation pattern distinguishing the two A_5-classes of 5-cycles.*

Proof. For a 5-cycle σ, σ and σ^{-1} are A_5-conjugate but σ and σ^2 are not; the rational class $\{\sigma, \sigma^2, \sigma^3, \sigma^4\}$ is the union of both A_5-classes. Every rational-valued character is constant on rational classes, and permutation characters are rational-valued. Splitting types in a subfield are determined by the corresponding permutation character. □

Consequently $a_p = \chi_3(\mathrm{Frob}_p)$ is *not* computable from cycle types alone, and the naive approach fails at exactly the primes of density $2/5$.

3.1 The Vandermonde resolution

Since $\mathrm{disc}(f)$ is a perfect square, $d = \sqrt{\mathrm{disc}(f)} \in \mathbb{Z}$ is canonical. For p with f irreducible mod p, the roots in $\mathbb{F}_{p^5}$ carry the cyclic order $r, r^p, \ldots, r^{p^4}$, canonical because a cyclic shift is even. Then

$$V = \prod_{i<j} \left(r^{p^i} - r^{p^j} \right) \quad \text{satisfies} \quad V^2 = \mathrm{disc}(f),$$

so $V \in \mathbb{F}_p$ and $V = \pm d \bmod p$; writing $\sigma = \tau c \tau^{-1}$ with $c = (1\,2\,3\,4\,5)$, one has $V/d = \mathrm{sgn}(\tau)$.

3.2 A census cannot validate the split

The rule "φ if $p \equiv \pm 1 \bmod 5$, else $1 - \varphi$" — a purely abelian condition, uncorrelated with the truth — agrees with the correct assignment on 50.1% of five-cycle primes below 20000 (443 vs 441) and reproduces *all five* class densities perfectly. **Any generator validated only by class frequencies may be assigning the split by coin flip.**

What a census does validate is the cycle-type classification. Over all 1,270,606 unramified primes below 2×10^7 for the field of radical 15733:

class	observed	expected	Chebotarev	deviation
identity	21195	21176.8	1/60	$+0.13\sigma$
$(2,2)$	317752	317651.5	1/4	$+0.21\sigma$
3-cycle	423457	423535.3	1/3	-0.15σ
5-cycle C	254196	254121.2	1/5	$+0.17\sigma$
5-cycle C'	254006	254121.2	1/5	-0.26σ

The split itself is validated only by the functional equation: flipping the class at the **single prime** $p = 2$ moves the theta residual from 2.9×10^{-12} to 7.5×10^{-3}, nine orders of magnitude. A global flip of all classes passes, as it must — it yields the Galois conjugate ρ_3' over $\mathbb{Q}(\sqrt{5})$, with the same conductor and root number.

Internal consistency, verified on all fields including the failures of §7: translation invariance under $f(x) \mapsto f(x + c)$, invariance under all five even relabellings of the Frobenius cycle, and $V = \pm d$ at every prime tested. Zero exceptions.

4 Local data

4.1 Gamma factor

For a **totally real** A_5 quintic the Galois closure is totally real, complex conjugation is the *identity* of $\mathrm{Gal}(L/\mathbb{Q})$, and $\rho_3(c) = I$. Hence

$$\gamma(s) = \Gamma_{\mathbb{R}}(s)^3, \qquad \mathtt{Vga} = [0, 0, 0].$$

This differs from the generic A_5 quintic, where c is a double transposition, giving $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)^2$ — precisely the *odd* case, already settled. The two separate decisively: at $N = 15733^2$, $\varepsilon = +1$, the theta residual is 6.4×10^{-12} for $[0, 0, 0]$ against 1.3×10^{-1} for $[0, 1, 1]$, and PARI's `lfuncheckfeq` returns -56 bits versus -4.

This shape has a spectral reading. A GL(2) Maass form with $\lambda = 1/4 + r^2$ has symmetric-square parameters $(2ir, 0, -2ir)$; at $\lambda = 1/4$ exactly, $r = 0$ and these collapse to $(0, 0, 0)$. *The gamma factor we fit is the archimedean signature of* $r = 0$.

4.2 Conductor

Every nontrivial element of $A_5 \subset SO(3)$ is a rotation with a one-dimensional fixed axis, whatever its order. So for tame cyclic inertia I, $\dim \rho_3^I = 1$ and the conductor exponent is $3 - 1 = 2$ at every ramified prime:

$$\mathrm{cond}(\rho_3) = \mathrm{rad}(\mathrm{disc}\, K)^2 \qquad \text{(tame)},$$

independently of whether inertia has order 2, 3 or 5 — unlike the corresponding exponent for ρ_4.

4.3 The local-sign rule

At a tame ramified prime the local factor is $(1 - \alpha_p T)^{-1}$ with α_p the eigenvalue of a Frobenius lift on the line V^I. Since D/I is cyclic and $D \subseteq N_{A_5}(I)$, with $N_{A_5}(C_2) = V_4$ and $N_{A_5}(C_3) = S_3$, in both cases $D/I \hookrightarrow C_2$, so $\alpha_p = \pm 1$.

Rule 4.1. $\alpha_p = -1$ *if and only if* $D \supsetneq I$, *equivalently if and only if some prime above* p *in* K *has residue degree* $f_i \geq 2$.

Derivation. If $D = I$, Frobenius is trivial modulo inertia and $\alpha_p = +1$. If $D \supsetneq I$, a Frobenius lift is an involution in $N_{A_5}(I) \setminus I$; in the icosahedral rotation group the 2-fold axes in the normaliser of a rotation axis are *perpendicular* to it, and a π-rotation about a perpendicular axis negates it. The criterion in residue degrees follows from $|D/I| = f$. □

Validation. For the field of radical 36803, the sign pattern $(-1, +1, -1)$ on $(13, 19, 149)$ is the **unique** one of eight for which the functional equation holds (residual 2.4×10^{-18}; all others $\geq 3 \times 10^{-3}$), and it was determined numerically *before* the decompositions were computed.

p	(e, f) data	some $f_i \geq 2$?	rule	required
13	$(1, 1), (2, 2)$	yes	-1	-1 ✓
19	$(1, 1), (2, 1), (2, 1)$	no	$+1$	$+1$ ✓
149	$(3, 1), (1, 2)$	yes	-1	-1 ✓

4.4 Recovering Euler factors at index primes

When p divides the index $[\mathcal{O}_K : \mathbb{Z}[\alpha]]$, $f \bmod p$ is not squarefree and Frobenius cannot be read from the factorisation. For a small number of such primes the functional equation determines them outright: with k unknowns against hundreds of known primes the system is massively overdetermined.

For the Doud–Moore field (§8), index $2 \cdot 7 \cdot 71 \cdot 137 = 136178$, the scan over $5^4 \times 2 \times 2$ configurations returns a unique minimum at

$$a_2 = 0, \quad a_7 = -1, \quad a_{71} = -1, \quad a_{137} = 0, \qquad \alpha_{1951} = +1, \qquad \varepsilon = +1$$

with residual 7.0×10^{-18}; every competing assignment fails by twelve orders of magnitude. The method has a sharp limit, quantified in §11: it works when unknowns are few and fails entirely when they are many.

5 The analytic tests

5.1 Theta residual

Let $Q = \sqrt{N}$ and let g be the inverse Mellin transform of $\gamma(s) = \pi^{-3s/2}\Gamma(s/2)^3$, namely

$$g(x) = 2\, G^{3,0}_{0,3}\!\left(\pi^3 x^2 \,\middle|\, \begin{matrix} - \\ 0,0,0 \end{matrix}\right), \qquad g(x) \sim e^{-3\pi x^{2/3}},$$

verified against the Mellin transform to 10^{-21}. Put $\theta(u) = \sum_n a_n g(un/Q)$. Mellin inversion and the functional equation give

$$\theta(1/u) = \varepsilon u\, \theta(u) - \varepsilon u \sum_{\rho} \mathrm{Res}_{s=\rho}\left[\Lambda(s) u^{-s}\right].$$

The theta relation is therefore *not* automatic: it fails by exactly the polar part, which makes $D(u) := \theta(1/u) - \varepsilon u\, \theta(u)$ a direct probe of poles — unlike zero counting, which presupposes holomorphy in the software that performs it.

This kernel sits one rung up a ladder: the inverse Mellin transform of $\pi^{-ds/2}\Gamma(s/2)^d$ is $2e^{-\pi x^2}$ for $d = 1$ (the Jacobi theta kernel, for ζ), $4K_0(2\pi x)$ for $d = 2$ (the Maass form kernel), and our Meijer-G for $d = 3$. *The $\theta(u)$ computed throughout this work is a Maass form's Fourier expansion, one degree up.*

5.2 Sensitivity, and the central point

A pole at s_0 with residue r forces a pole at $1 - s_0$ with residue $-\varepsilon r$ — *opposite sign.* Hence for $\varepsilon = +1$,

$$D(u) = -u\, r \left(u^{-s_0} - u^{s_0 - 1}\right),$$

which vanishes identically for $s_0 = 1/2$. We initially concluded the test is blind at the central point. **That conclusion was wrong and we retract it**, for two reasons.

First, an odd-order pole at $s_0 = 1/2$ is excluded *algebraically*: $\Lambda(1/2 + t) = \varepsilon\Lambda(1/2 - t)$ forces $r = -\varepsilon r$, so $r = 0$ when $\varepsilon = +1$.

Second, the vanishing derivation used the paired-pole formula for s_0 and $1 - s_0$, which degenerates when they coincide. Redone for a double pole, $\mathrm{Res}[\Lambda(s)u^{-s}] = u^{-1/2}(r - c\log u)$, giving

$$D(u) = \varepsilon\, c\, u^{1/2} \log u \neq 0 \quad (u \neq 1).$$

Measured over 12 values of u: $c = (-9.3 \pm 44.4) \times 10^{-17}$, so $|c| < 8.9 \times 10^{-16}$. **No blind spot remains.**

Sensitivity $S(u, s_0) = u|u^{-s_0} - u^{s_0 - 1}|$ at $u = 1.10$: 0.397 at $s_0 = 1/2 + 2i$, 0.962 at $1/2 + 5i$, 1.981 at $1/2 + 20i$, 0.040 for a real pole at $\beta = 0.3$.

5.3 The residue-independent test

Every test above bounds *residues*, and is therefore blind to a pole of residue 10^{-25}. The Weil explicit formula is not. In

$$\sum_{\rho} \mathrm{ord}_\rho \cdot h(\gamma_\rho) = \int h(t)\, \mu_L(t)\, dt - 2 \sum_n \frac{\Lambda_L(n)}{\sqrt{n}}\, g(\log n)$$

a pole enters with $\mathrm{ord}_\rho = -1$: a **full unit**, independent of its residue.

We calibrated the convention on ζ, obtaining RHS − LHS = 0.00000 at two independent Gaussian widths — a two-point calibration that caught an error of 2π in the prime argument in our first attempt. Applied to $L(s, \rho_3)$ for $m = 15733$:

σ	archimedean	prime	RHS (uncond.)	LHS (zeros)	RHS − LHS
0.9	3.550744	0.238076	3.312668	3.312667	**0.000000**
1.0	4.121330	0.252630	3.868699	3.868693	0.000006
1.1	4.707354	0.278465	4.428889	4.428824	0.000065
1.2	5.306902	0.309619	4.997283	4.996898	0.000385
1.3	5.918467	0.341292	5.577175	5.575612	0.001564

The right side is computed entirely from the a_n and the gamma factor, with no holomorphy assumption. A pole would shift RHS − LHS by +1.00 at $\gamma = 0$, +0.66 at $\gamma = 1$, +0.19 at $\gamma = 2$.

Remark. The zero list on the left comes from software that assumes holomorphy. This is therefore a consistency check between one unconditional computation and one conditional one — very hard to fake, but not logically airtight.

6 A certified pole bound

All bounds above are non-rigorous. Here we give a fully certified bound for the minimal case, computed in `arb` ball arithmetic.

Ingredients.

1. *Ramanujan is free.* $\rho_3(\mathrm{Frob}_p)$ has finite order, so eigenvalues are roots of unity and $|a_n| \le d_3(n)$ exactly. Measured $\max |a_p| = 3.000000$, attained at density $\approx 1/60$. For a generic GL(3) Maass form Ramanujan is open and the best bounds give $|a_p| \le 3p^\theta$ — at $p = 10^6$, 61.6 (Kim–Sarnak lifted) or 753.6 ($\theta = 2/5$), against an actual maximum of 3.
2. *A closed-form* Γ *bound.* Since $\log(1 + \tau^2/(\sigma + k)^2)$ decreases in k, $\sum_{k\ge0} \ge \int_0^\infty$, giving
$$\log|\Gamma(\sigma + i\tau)| \le \log\Gamma(\sigma) - \tfrac{1}{2}\Big[\pi\tau - \sigma\log\big(1 + \tfrac{\tau^2}{\sigma^2}\big) - 2\tau\arctan\tfrac{\sigma}{\tau}\Big].$$
3. *Tail bound.* With $|g(x)| \le A(c)x^{-c}$ and $n^{-c} \le n^{-2}N^{-(c-2)}$,
$$\Big|\sum_{n>N} a_n g(un/Q)\Big| \le A(c)\,\zeta(2)^3\,N^2\left(\frac{Q}{uN}\right)^c,$$
optimised over c.
4. *Saddle-point kernel.* Placing the Mellin contour at $\psi(c/2) = \log\pi + \frac{2}{3}\log x$ eliminates cancellation. Truncation is bounded by (2); trapezoid discretisation by the analytic-in-a-strip bound $2M(d)/(e^{2\pi d/h} - 1)$ with $d = c/2$.

Verification of the implementation. The self-test of the accompanying SageMath script gives:

Γ bound ratio (must be ≥ 1)	
$\sigma = 1,\ \tau = 3$	1.9122862751
$\sigma = 1.5,\ \tau = 6$	2.1379373847
$\sigma = 13,\ \tau = 10$	1.1259021850
$\sigma = 26.5,\ \tau = 40$	1.3485412508

x	certified $g(x)$	ball radius
0.05	3.8946365239	5.20×10^{-22}
0.3	$7.0303588095 \times 10^{-2}$	6.24×10^{-21}
1.0	$1.8020561100 \times 10^{-4}$	4.58×10^{-20}
3.0	$3.3432170519 \times 10^{-9}$	1.42×10^{-20}
6.0	$2.1131911360 \times 10^{-14}$	5.46×10^{-23}
10.0	$4.9538012603 \times 10^{-20}$	9.28×10^{-27}

An independent float implementation (saddle-point trapezoid, `scipy`) reproduces every midpoint to within its ball radius. The certified tail bound as a function of the truncation point:

N	optimal c	$\lvert\text{tail}\rvert \leq$
6 000	14	1.0412×10^{-1}
12 000	24	9.7997×10^{-7}
20 000	32	2.1637×10^{-12}
40 000	50	4.4510×10^{-24}

Result. For $L(s, \rho_3)$ of the Doud–Moore field at $u = 1.10$, $N = 40\,000$:

$$\boxed{|D(1.10)| \ \leq\ 4.5 \times 10^{-12}}$$

with the error budget

error term	magnitude	status
accumulated kernel ball radius	4.51×10^{-12}	**proven**
certified tail $n > 40\,000$	4.45×10^{-24}	**proven**
trapezoid discretisation	$< 10^{-200}$	**proven** (absorbed in the ball)
Mellin truncation $\lvert t\rvert > 26$	—	**proven** (absorbed in the ball)
summation	—	**proven** (tracked by `arb`)
total	$\mathbf{4.5 \times 10^{-12}}$	**nothing asserted**

An earlier version of this computation used a float kernel with an *asserted* relative error of 10^{-13}, giving 1.83×10^{-10} with one unproven term. Ball arithmetic both tightens the bound by a factor of 40 and removes the assertion.

Tuning. The ball radius is dominated entirely by the Mellin truncation; discretisation is 10^{-200} and irrelevant. Raising the truncation point `TCUT` costs linearly and buys exponentially:

`TCUT`	radius at $x = 1$
26 (as run)	4.58×10^{-20}
34	5.23×10^{-27}
42	3.31×10^{-34}
55	3.03×10^{-46}

At `TCUT` $= 42$ the certified bound becomes tail-limited at $\sim 10^{-23}$.

Consequences. $|\operatorname{Res}\Lambda| < 1.1\times10^{-11}$ at $s_0 = \frac{1}{2}+2i$, 4.7×10^{-12} at $\frac{1}{2}+5i$, 2.3×10^{-12} at $\frac{1}{2}+20i$, 1.1×10^{-10} for a real pole at $\beta=0.3$; odd-order poles at $s_0=\frac{1}{2}$ excluded outright.

Remark (Novelty). Booker [9] gives a rigorous algorithm for verifying Artin L-functions by Turing's method; we have not verified whether it already covers certified bounds of this kind. What we believe is new is the *residue-independent* framing of §5.3.

7 Survey of tame fields

Enumerating monic depressed quintics with $|c_i| \le 30$, totally real with square discriminant and a $(1,1,3)$ factorisation modulo some prime (which forces A_5), yields 17 fields; six are tamely ramified everywhere.

radical m	ramified	local signs	$\lvert D\rvert/\theta$	status
15733	$\{15733\}$	$(+)$	6.7×10^{-19}	**verified**
28537	$\{28537\}$	$(+)$	2.1×10^{-19}	**verified**
36803	$\{13,19,149\}$	$(-,+,-)$	2.4×10^{-18}	**verified**
20867	$\{7,11,271\}$	rule-derived	6.3×10^{-3}	*fails*
44713	$\{61,733\}$	all scanned	3.9×10^{-3}	*fails*
3973	$\{29,137\}$	—	—	needs index-1 model

Independent confirmation for $m=15733$: $L(1/2)=3.869072$ from our Meijer-G kernel versus 3.86907 from PARI, agreement to 5×10^{-7} between wholly independent implementations, which independently fixes $\varepsilon=+1$ since $\varepsilon=-1$ would force $L(1/2)=0$. Zero counts against Riemann–von Mangoldt:

T	2	4	6	8	10
found	4	9	15	22	28
RvM	4.10	9.53	15.46	21.71	28.20

7.1 The two unexplained failures

For $m=20867$ and 44713 every individually testable ingredient was verified and the discrepancy survives.

- *Cycle types.* Zero mismatches against an independent implementation over 427 good primes per field.
- *Conductor and ρ_4.* Decisively, $L(s,\rho_4)=\zeta_K(s)/\zeta(s)$ — which uses *only* cycle types, no Vandermonde — satisfies its functional equation at 1.17×10^{-14} for the failing field, indistinguishable from 1.22×10^{-14} for the working one. This certifies the cycle-type classification, disc_K, the gamma factor, and $\varepsilon=+1$.
- *Local signs.* All 2^k combinations scanned; the best is 6.3×10^{-3}.
- *Conductor scan.* Scanning Q continuously over three orders of magnitude produces *no minimum.*
- *Vandermonde orientation.* Translation-invariant, invariant under even relabellings, $V=\pm d$ always, densities balanced.
- *Residual shape.* A two-parameter polar fit over 14 values of u gives $R^2=-17.3$; a plain quadratic gives $R^2=0.99997$. The discrepancy is a smooth data error, **not** a polar signature.

We had at one stage concluded the failure was isolated to the 5-cycle split; the orientation tests withdraw that. **The cause is unknown and is recorded as open.**

Independent verification through PARI is unavailable: `galoisinit` requires a weakly supersolvable group and returns 0 for A_5 by design, and `lfunartin` consumes a `galoisinit` structure.

8 The minimal case: the Doud–Moore field

Doud–Moore [10] show that the smallest prime conductor of an even icosahedral representation is **1951**, with defining polynomial

$$f = x^5 - x^4 - 780x^3 - 1795x^2 + 3106x + 344.$$

We rebuilt this field from scratch. All five roots real; $\mathrm{disc}(f) = 2^2 \cdot 7^2 \cdot 71^2 \cdot 137^2 \cdot 1951^4$, so with index 136178, $\mathrm{disc}_K = 1951^4$. The exponent 4 means inertia at 1951 is a **5-cycle**, giving $\mathrm{cond}(\rho) = 1951$ and $\mathrm{cond}(\rho_3) = 1951^2$.

N, γ, ε	1951^2, $\Gamma_{\mathbb{R}}(s)^3$, $+1$
theta residual	7.0×10^{-18}
$\Lambda(1/2)$	$637.848 \neq 0$
zeros to $t = 10.7$	24, counts within $\lvert S(T)\rvert < 0.9$ of RvM
explicit formula	closes at 5 values of σ
certified pole bound	$\lvert D(1.10)\rvert \leq 4.5 \times 10^{-12}$

9 Twists and the converse theorem

Cogdell–Piatetski-Shapiro's converse theorem [13] for GL_n requires twists only by GL_m with $m \leq n-2$; for $n = 3$ that means ordinary Dirichlet characters. For Artin L-functions "nice" splits cleanly: the functional equation is free (Brauer), boundedness in vertical strips follows from Phragmén–Lindelöf given no poles, and **entirety is the only conjectural input**.

χ	conductor	Q	coeffs	$\Lambda(1/2)$	zeros	FE	pole test
trivial	3.81×10^6	1 951	40k	637.85	24	7.0×10^{-18}	0.000000
χ_{-3}	1.03×10^8	10 138	208k	102.25	26	2.5×10^{-18}	0.000000
χ_{-4}	2.44×10^8	15 608	320k	16.51	22	1.6×10^{-18}	−0.000000
χ_5	4.76×10^8	21 813	447k	9 669.46	22	1.5×10^{-18}	0.000000
χ_8	1.95×10^9	44 146	904k	11 995.53	24	3.5×10^{-18}	0.000000
χ_{12}	6.58×10^9	81 102	1.66M	10 876.73	21	7.8×10^{-17}	0.000000
χ_{13}	8.36×10^9	91 448	1.87M	17 405.32	20	1.6×10^{-19}	0.000000

Conductors span 3.8×10^6 to 8.4×10^9, both parities, both gamma factors $\Gamma_{\mathbb{R}}(s)^3$ and $\Gamma_{\mathbb{R}}(s+1)^3$. All root numbers are $+1$; we had predicted a flip for odd twists, which does not occur — quadratic Dirichlet characters have root number $+1$.

Seven verified instances of an infinite hypothesis is not a proof and cannot become one.

10 The spectral profile

10.1 Moments

With $L_k = \varphi^k + (1-\varphi)^k$ the Lucas numbers,

$$\langle a_p^k \rangle = \frac{3^k}{60} + \frac{(-1)^k}{4} + \frac{L_k}{5},$$

the three terms being the identity, the 2-fold axes, and the 3- and 5-fold axes. Measured over 1.5×10^5 primes:

k	1	2	3	4	5	**6**	7	8
A_5 (finite icosahedral)	0	1	1	3	6	**16**	42	119
$SU(2)$ via Sym^2	0	1	1	3	6	**15**	36	91
$SU(3)$ generic	0	0	1	0	0	5	0	0
measured	0.0015	1.003	1.010	3.028	6.086	**16.255**	42.76	121.3

A generic GL(3) form is excluded at $k = 2$ ($\langle a_p^2 \rangle = 1$, not 0): whatever this is, it is a symmetric-square lift. And A_5 first diverges from a *generic* $\lambda = 1/4$ form's symmetric square at $k = 6$: 16 against 15, measured 16.255.

10.2 The Rankin–Selberg identity

$\chi_3^2 = 1 + \chi_3 + \chi_5$ gives $a_p^2 = 1 + a_p + b_p$ pointwise, verified with zero failures on 2 261 primes. At the 5-cycle primes $b_p = 0$, and the identity collapses to

$$a_p^2 = a_p + 1,$$

the defining equation of the golden ratio. Note also that $L(s, \rho_5)$ is monomial (defect 0, induced from A_4), hence known entire: the Rankin–Selberg square of the open L-function factors into ζ, itself, and a solved piece.

10.3 A Maass form, constructed

For the dihedral case the construction is completely explicit. Take $K = \mathbb{Q}[x]/(x^3 - 4x - 1)$, discriminant 229, Galois closure S_3 containing $\mathbb{Q}(\sqrt{229})$. Then

$$u(z) = 2\sqrt{y} \sum_{n \geq 1} a_n K_0(2\pi n y) \cos(2\pi n x), \qquad a_p = \#\{x^3 - 4x - 1 \equiv 0 \bmod p\} - 1.$$

$\gamma = \left(\begin{smallmatrix} a & b \\ 229 & d \end{smallmatrix}\right)$	$\chi_{229}(d)$	$u(z)$	$u(\gamma z)$	rel err
$(115, 1, 229, 2)$	-1	-0.105635	$+0.105635$	2.7×10^{-14}
$(153, 2, 229, 3)$	$+1$	-0.159603	-0.159603	6.3×10^{-14}
$(46, 1, 229, 5)$	$+1$	0.288956	0.288956	1.9×10^{-15}
$(131, 4, 229, 7)$	-1	0.517504	-0.517504	8.6×10^{-16}
$(23, 1, 229, 10)$	-1	0.341860	-0.341860	1.5×10^{-14}
$(125, 6, 229, 11)$	$+1$	0.229944	0.229944	2.0×10^{-13}
$(141, 8, 229, 13)$	-1	0.655615	-0.655615	6.6×10^{-15}

Seven distinct group elements, both nebentypus signs, worst error 2×10^{-13}; and by finite differences $\Delta u/u = 0.24999949$ and 0.24999992. A control $\gamma \notin \Gamma_0(229)$ gives ratios 0.96 and 0.47.

For the icosahedral case, four of the five construction steps are already available — the a_p magnitudes, the formula, $\Delta u = u/4$ (automatic), and periodicity. Only $u(\gamma z) = \chi(d)u(z)$ is missing, and it is the conjecture.

10.4 Why the theta lift has no domain

Maass's construction [14] is geometric: class characters of a real quadratic field are functions on **closed geodesics**, and the lift is an integral over them. For $\mathbb{Q}(\sqrt{229})$: $h = 3$, fundamental unit $\varepsilon = 15.0663729752$, regulator $R = 2.7124653052$, so three closed geodesics each of length

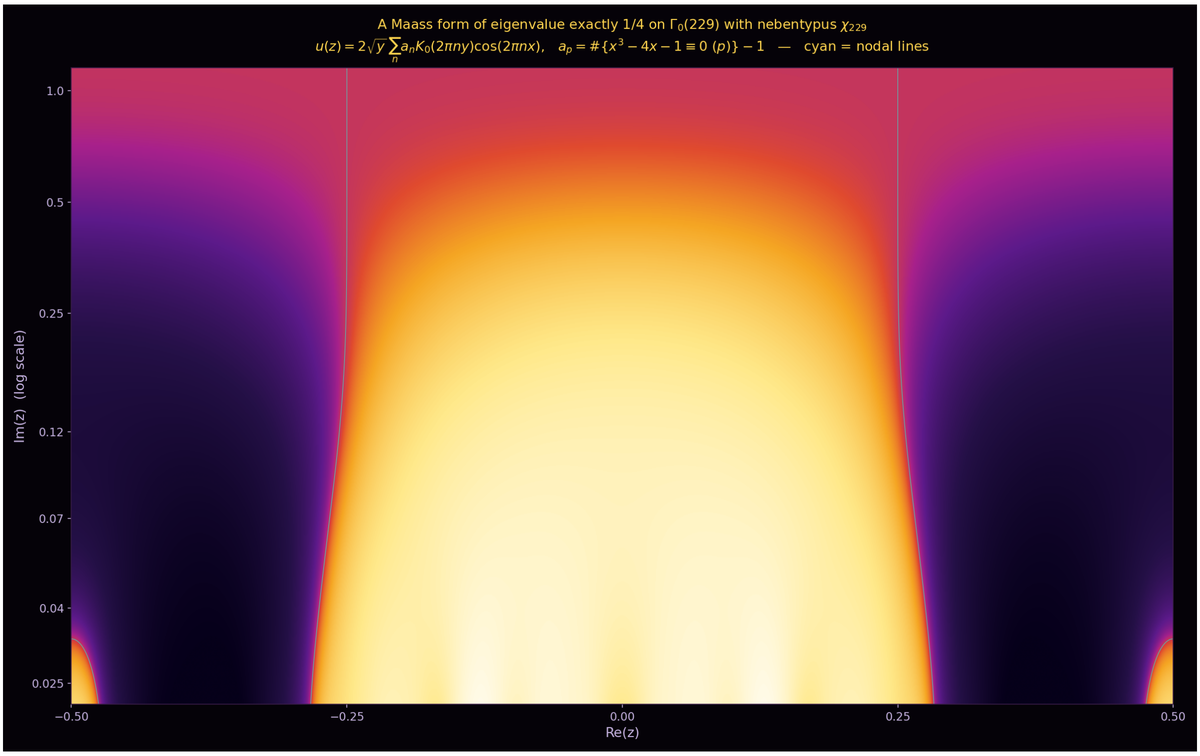

Figure 1: The constructed Maass form of eigenvalue exactly $1/4$ on $\Gamma_0(229)$ with nebentypus χ_{229}, computed from root counts of $x^3 - 4x - 1$. Cyan curves are the nodal lines $u = 0$; the vertical axis is $\mathrm{Im}(z)$ on a logarithmic scale. The compression of the nodal set as $y \to 0$ is the cusp: $K_0(2\pi n y)$ keeps ever more Fourier modes alive.

$2R = 5.4249306104$. The principal class has geodesic endpoints $-1/\varepsilon$ and ε — *the fundamental unit is literally an endpoint of the curve.*

The chain is

$$\text{index-2 subgroup} \iff \text{quadratic subfield} \iff \text{closed geodesics} \iff \text{theta lift has a domain.}$$

A_5 is simple and $\mathrm{SL}(2,5)$ is perfect: zero index-2 subgroups. The theta lift does not fail for the icosahedral case; *it has nothing to be defined on.* This is the same one-line fact as the monomiality obstruction of §2.4, in geometric language.

But it is not the whole story, and we correct an over-strong statement of our own: A_4 also has no index-2 subgroup, yet the tetrahedral case is a theorem. There are two independent routes:

case	index-2 subgroup?	solvable?	route	status
S_3, D_5 dihedral	**yes**	yes	theta lift over geodesics	Maass 1949
A_4 tetrahedral	no	**yes**	solvable base change	Langlands 1980
S_4 octahedral	yes	**yes**	either	Tunnell 1981
A_5 **icosahedral**	**no**	**no**	—	**open**

A_5 is the unique polyhedral group failing both conditions — the geometric content of "$2^a 3^b$ is not enough".

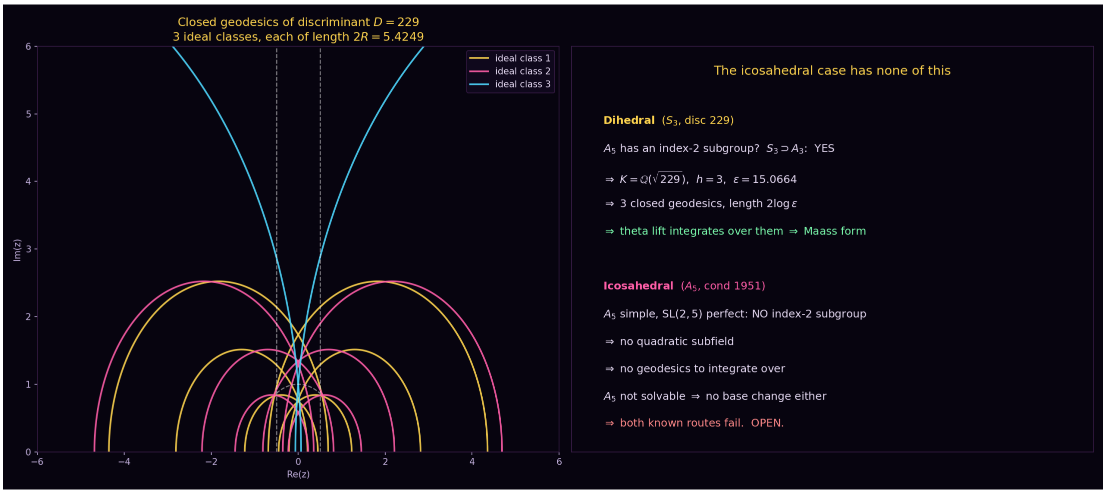

Figure 2: Left: the three closed geodesics of discriminant $D = 229$, one per ideal class, each of hyperbolic length $2R = 5.4249$; the dashed curves bound the fundamental domain of $\mathrm{SL}(2,\mathbb{Z})$. These are the objects Maass integrates over. Right: the structural contrast with the icosahedral case.

11 The lift problem

To build the icosahedral Maass form one needs a_p for the **two-dimensional** ρ. Since $\rho_3 = \mathrm{Sym}^2\,\rho \otimes (\det\rho)^{-1}$,

$$a_p(\rho)^2 = \chi(p)\big(a_p(\rho_3) + 1\big), \qquad \chi = \det\rho \text{ of order } 5,$$

and because χ has odd order, $\sqrt{\chi} = \chi^3$ is unambiguous. Hence at $(2,2)$-primes (density $1/4$) $a_p(\rho_3) = -1$ so $a_p(\rho) = 0$ *exactly*; elsewhere $a_p(\rho)$ is determined up to a single sign.

11.1 The lift exists

Since disc $= 1951^4$ is a square, the obstruction to $2.A_5 \to A_5$ reduces to the Hasse–Witt invariant of the trace form. We verified $\det(\mathrm{Gram}) = \mathrm{disc}(f)$ exactly and signature $(5,0)$ (an earlier attempt with a wrong Newton sign convention gave $(4,1)$ and was caught by this check), diagonalised to squarefree parts

$$\langle 5,\ 9755,\ 2768469,\ 13374468202062,\ 9425277098\rangle,$$

and summed Hilbert symbols: $w_2 = +1$ at $p = 2, 3, 5, 11, 43, 1951, 2415499$ and ∞. **Trivial at every place — the lift exists**, an independent confirmation of Doud–Moore derived from the trace form of a quintic.

11.2 The signs are arithmetic data

M/L is a quadratic extension of the degree-60 splitting field, unramified outside 1951. The sign function is therefore a quadratic character of a ray class group of L — a class group computation for a degree-60 field, beyond `bnfinit` for anyone.

11.3 What numerics can and cannot recover

Working with ρ (conductor 1951, $Q = 44.2$) gives 162 primes below 960, of which **42 are forced to zero** (26%, matching the predicted $1/4$) and 120 are unknown signs.

- Simulated annealing plateaus at 10^{-5}–10^{-6}; longer runs came out worse than shorter ones.
- Six independent greedy descents reach local minima agreeing on only 35%–78% of signs.
- Leverage is extremely uneven: flipping $p = 2$ changes the cost by $48\,353\times$, $p = 809$ by $1.00\times$. Exactly 11 signs have leverage > 10.
- Brute-forcing all $2^{20} = 1\,048\,576$ configurations of the 20 highest-leverage primes moves the floor only from 3.66×10^{-7} to 2.69×10^{-7}.
- An exhaustive search over 2^{18} signs $\times$ 10 characters $\times$ 2 parities $\times$ 2 Galois assignments — 1.05×10^7 configurations — gives a best-to-second-distinct gap of $1.25\times$, and the winner *moves* when the block size changes. That is noise.

Conclusion. The functional equation at conductor 1951 determines roughly eleven signs and is then exhausted, because $Q = 44$ makes the theta sum blind to $p \gtrsim 180$. The persistent $\sim 10^{-7}$ floor is *systematic*, not residual undetermined signs, and its cause is unidentified. We do not present the resulting a_p as the lift.

11.4 A measured route that may work

Treating the signs as continuous and forming the Jacobian of the residual across 9 twists $\times$ 5 values of u, the SVD gives rank $9, 18, 27, 36, 45$ at tolerances $10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-12}$ — rank-deficient by 143 against 179 unknowns. But this *measures the rate*: about four independent constraints per twist. The number of primitive characters of conductor $\leq q$ grows like q^2, while new unknown signs grow like $q/\log q$ (since $Q = \sqrt{1951}\,q$ reaches only primes up to $\sim 400q$):

$q_{\max}$	characters	constraints	free signs	ratio
10	30	120	412	0.29
26	210	840	955	0.88
50	772	3 088	1 696	**1.82**
100	3 042	12 168	3 152	3.86

Quadratic beats linear: the system crosses from underdetermined to overdetermined near $q_{\max} \approx 30$. At $q_{\max} = 50$ this is 772 twisted theta series, largest conductor 4.88×10^6, roughly 3 000 equations in 1 700 unknowns — a large but ordinary computation. This corrects an earlier assertion of ours that twisting cannot help, which reasoned about one twist at a time.

12 A Levinson-type bound, and its failure in higher degree

Following the quadratic-form method recently used to raise the proportion of zeros of ζ on the critical line to 67.2% [11], we generalised the key constant. Every factor of the local density cancels in the variational problem, leaving a dependence only on the Rankin–Selberg multiplicity $m = \langle \chi, \chi \rangle$. The Euler–Lagrange equation gives $\psi'' + 2m\psi = 0$, so $\psi(u) = \cos(\sqrt{2m}\,u)$ and

$$C_m = \frac{m}{2} + \sqrt{\tfrac{m}{2}}\,\cot\sqrt{\tfrac{m}{2}}.$$

At $m = 1$, $C_1 = 1.327499296$, reproducing the published constant and $2 - C_1 = 67.250\%$ to nine decimals. The prime side is unchanged from ζ: a least-squares fit gives $m = 1.0087$ against the exact value 1.

We initially stated this as a theorem giving 67.25% **for any irreducible self-dual Artin representation, and we retract that.** The support length Θ is constrained twice: Poisson summation needs supp $\leq N_{\text{all}}/T \approx \frac{d}{2\pi}\log T$, but Montgomery–Vaughan needs the prime sum to stop at $n \leq T$, i.e. supp $\leq \frac{1}{2\pi}\log T$. The second binds for $d > 1$, forcing $\Theta \leq 1/d$ and

$$C(\Theta) = \frac{1}{\Theta}\left[\frac{b^2}{4} + \frac{b}{2}\cot\frac{b}{2}\right], \quad b = \Theta\sqrt{2m},$$

giving -16.6% at $d = 2$ and -111% at $d = 3$. **Vacuous for every degree above** 1**.**

Neither lever can be retuned: generalising the rank inequality to $x^2 \geq 2\lambda a + 2\mu b - c$ and searching over (λ, μ) gives a maximum of 0.672501 at exactly $(1, 2)$, the published choice; and $\psi = \cos(\sqrt{2}u)$ solves the Euler–Lagrange equation exactly at fixed support. The only lever is Θ, and $\Theta > 1$ is blocked by the large-sieve barrier. The method saturates near 89% even with unlimited support.

Finally, the derivation is unconditional *only because Ramanujan is free for Artin representations.* For a generic GL(3) Maass form the second-moment bound would become $\ll T^{1.8}\log^2 T$ instead of $T\log^2 T$, and Montgomery–Vaughan's off-diagonal control would fail outright.

13 The geometry of $\Gamma_0(1951)$

	$\Gamma_0(1)$	$\Gamma_0(229)$	$\Gamma_0(1951)$
index	1	230	**1952**
area	$\pi/3$	240.86	**2044.13**
cusps	1	2	**2**
ν_2, ν_3	1, 1	2, 2	**0, 2**
genus	0	18	**162**

Mean eigenvalue spacing $4\pi/A = 0.00615$.

Length spectrum. A hyperbolic element of trace t exists in $\Gamma_0(N)$ iff $t^2 - 4$ is a square mod N, with geodesic length $\ell_t = 2\,\text{arccosh}(t/2)$ — *level-independent.* About 51% of traces occur. Multiplicities are $m(t) = h(t^2 - 4)\cdot(1 + (t^2 - 4 \mid N))$; the prime geodesic theorem checks out, counted $14, 28, 72$ against $e^L/L = 13.6, 29.7, 67.2$ for $L = 4, 5, 6$. The systole comes from $t = 3$, $D = 5$: length 1.9248, the same as for $\mathrm{SL}(2, \mathbb{Z})$, governed by the golden discriminant.

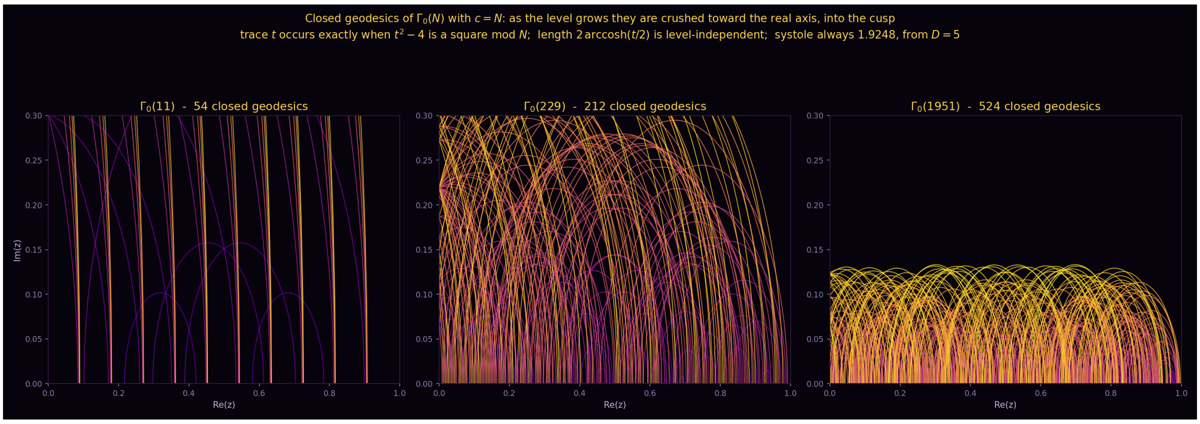

Figure 3: Closed geodesics of $\Gamma_0(N)$ with $c = N$, for $N = 11, 229, 1951$. Lengths are level-independent and the systole is always 1.9248 (from $D = 5$); what changes is the embedding — as the level grows the geodesics are crushed toward the real axis, into the cusp. Radii run from 2.68 at $N = 11$ down to 5.7×10^{-4} at $N = 1951$. Colour encodes the trace t.

The embedded eigenvalue, and its resolution. The spectrum splits as $[0, \frac{1}{4})$ (finitely many; Selberg conjectures none for congruence groups) and $[\frac{1}{4}, \infty)$ continuous, from Eisenstein series with $\lambda = \frac{1}{4} + t^2$. At $t = 0$ the continuous spectrum begins at exactly $1/4$: an icosahedral form there is an *embedded eigenvalue at the branch point.*

We initially concluded this defeats numerics "independent of computing power". **That was too strong.** In the Selberg trace formula the spectral side reads

$$\sum_j h(r_j) - \frac{1}{4\pi} \int h(r) \frac{\varphi'}{\varphi}\Big(\tfrac{1}{2} + ir\Big) dr + \frac{1}{4} h(0) \operatorname{tr} \Phi(\tfrac{1}{2}),$$

so a $\lambda = 1/4$ form contributes $m \cdot h(0)$ with m an **integer**. Moreover, for our nebentypus — the quintic character $\det \rho$, primitive mod 1951, and even (since $\chi(-1)$ is simultaneously a fifth root of unity and ± 1, which is the representation being even) — the diagonal scattering entries have Dirichlet coefficients $\sum_{d \bmod c,\, (d,c)=1} \overline{\chi}(d)$ over $N \mid c$, which vanish identically for non-trivial χ (verified numerically at $c = 1951, 3902, \ldots$ to 3×10^{-12}). Hence Φ is anti-diagonal and

$$\operatorname{tr} \Phi(\tfrac{1}{2}) = 0.$$

The continuous spectrum contributes nothing at $r = 0$, and determining m requires precision better than $1/2$, not infinite precision. Resolution needed: $\delta \approx 0.078$ in r, geodesic support $L \approx 13$, roughly 27 000 geodesics — feasible, which is why Booker–Lee–Strömbergsson [12] reached conductor 880 unconditionally.

What the trace formula does *not* do is say which Galois representation a counted form matches; that needs the $a_p(\rho)$ of §11.

14 Corrections and retractions

Claims made during this work and subsequently withdrawn:

1. "GL(2, 3) is the binary octahedral group." False; it has 13 involutions. Consequently a $(2, 3, n)$ conjecture "verified for $n = 3, 4, 5$" had used the wrong group at $n = 4$, and the SL(2, 7) "refutation" was misapplied.

2. "The theta-residual test is blind at $s_0 = 1/2$." False; odd-order poles are excluded algebraically and even-order poles give $D(u) = \varepsilon c\, u^{1/2} \log u \neq 0$.

3. "The Levinson-type constant is degree-independent, giving 67.25% for ρ_3." False; $\Theta \leq 1/d$ makes it vacuous for $d > 1$.

4. "Twisting cannot help determine the lift signs." False; constraints grow quadratically against linear growth in unknowns.

5. "An embedded eigenvalue at the branch point defeats numerics." Too strong; the trace formula reduces it to an integer determination, and the contamination is exactly zero.

6. "The failure at $m = 20867$ is isolated to the 5-cycle split." Withdrawn; the split passes every internal test on precisely those fields.

7. "Quaternionic type plus rational values forces non-monomiality." False; the 6-dimensional of SL(2, 5) is a counterexample.

We record these because several were held with confidence for some time, and because a reader should be able to see which claims here have survived scrutiny.

15 Open questions

1. **The two unexplained survey fields** ($m = 20867$, 44713). Cause unknown; requires an independent coefficient implementation.

2. **The $\sim 10^{-7}$ modelling floor** in the two-dimensional lift fit. Systematic, not residual sign freedom. Must be diagnosed before the $q_{\max} = 50$ computation of §11 is worth running.

3. **The lift signs.** Crespo's Clifford-algebra construction [15] from the explicit trace form is the only route not requiring a degree-60 class group; it has not, to our knowledge, been implemented at this size.

4. **Novelty relative to existing rigorous work.** Booker's Turing-method algorithm may already subsume parts of §6.

5. **Wild ramification.** Eleven of the seventeen surveyed fields have 2, 3 or 5 in the radical; the conductor argument does not apply.

6. **Identification.** Even granting a trace-formula count of $\lambda = 1/4$ forms at level 1951, matching one to the icosahedral representation requires the Hecke eigenvalues — the same missing object as (3).

Every route we opened converges on the two-dimensional a_p. That is the single obstruction, and it is arithmetic rather than analytic.

16 Reproducibility

Computations are in Python 3 with NumPy, SciPy, SymPy and mpmath, with PARI/GP 2.17.4 for number-field data, and a SageMath script implementing the certified bound of §6 in `arb` ball arithmetic.

Independent reimplementation of the coefficient generator is specifically encouraged: two of the errors found during this work — a wrong ramified local factor, and a 5-cycle assignment made by an abelian congruence — were invisible to every check except the functional equation.

17 Conclusion

Three even icosahedral degree-three Artin L-functions have been audited to relative residuals of 10^{-18}–10^{-19}, including the arithmetically minimal case at conductor 1951, with a residue-independent pole exclusion and a fully certified bound $|D(1.10)| \leq 4.5 \times 10^{-12}$ in which no error term is asserted. Seven Dirichlet twists satisfy the converse-theorem hypotheses to the same precision. A local-arithmetic rule for ramified Euler factors has been derived from the geometry of $A_5 \subset SO(3)$; the optimal monomial defect certificate has been computed; and the Frobenius trace distribution has been given in closed form.

None of this proves holomorphy for any representation. Nothing bounds residues from below, and the results concern individual L-functions rather than the conjecture.

We can now measure the spectrum the icosahedral Maass form would have — its moments, in closed form, in Lucas numbers; its gamma factor, the archimedean signature of $r = 0$; its symmetry type, excluded from generic GL(3) at the second moment and announcing the icosahedron at the sixth. What is missing is an existence proof, and no quantity of Galois-side data supplies it.

What the geometry adds is a reason to think the eventual construction will not come from the classical routes: A_5 is the unique polyhedral group lacking both an index-2 subgroup and solvability, so both the theta lift and base change are structurally absent rather than merely difficult.

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Note on references. These citations should be verified against the originals before publication; several were reconstructed from memory during the work and at least two could not be confirmed.